energy of the electrons is given in terms of the gas stream velocity U by

$$\tfrac{1}{2}m_e < v_r^2 > = \tfrac{1}{2}m_g U^2$$

Inelastic collisions will tend to reduce the electron energy, but this relationship shows that when U is near U_{crit} the electron energy is an appreciable fraction of eV_i and consequently ionization is to be expected. By using an equivalent field technique[5] which is closely related to the equivalent pressure technique of Blevin and Haydon[6], the electron ionization frequency can be found as a function of U alone from the data available on the electrons' drift velocity and ionization frequency in the absence of a magnetic field. The electron ionization frequency for helium is shown in Fig. 1, which also gives U_{crit} for helium.

The relative kinetic energies of interest measurements[7] show that the charge transfer collision cross-section for atomic helium ions in helium is much larger than the elastic cross-section, and at these energies elastic collisions show pronounced forward scattering[8]. For these reasons, charge transfer collisions are the most important, and it is a characteristic of these collisions that after the collision the ion has very nearly the same velocity, that is, U, as the atom had before it. The ion will then move through its Larmor orbit in the magnetic field with a constant velocity and the ion energy relative to the gas stream will then be $(m_i = m_g)$

$$m_g U^2 (1 - \cos \omega_i t)$$

The average ion energy is $m_g U^2$ and this equals the threshold energy for ionization ($2\ eV_i$) when $U = U_{crit}$. The peak energy is $2m_g U^2$, which is 98 eV in helium when $U = U_{crit}$, and these high energies can lead to appreciable ionization by the ions. The ionization frequency of the ions can be averaged around the orbits[5] of the ions and the result, based on Rostagni's[9] ionization cross-sections, is shown in Fig. 1. It can be seen that the ionization frequency of the ions is greater than that of the electrons although the ionization cross-section of the ions is much smaller than that of the electrons. This is because the energy of every ion is above threshold in part of its orbit, if U is large enough, while only the small fraction of electrons in the high energy tail of the electron energy distribution can ionize.

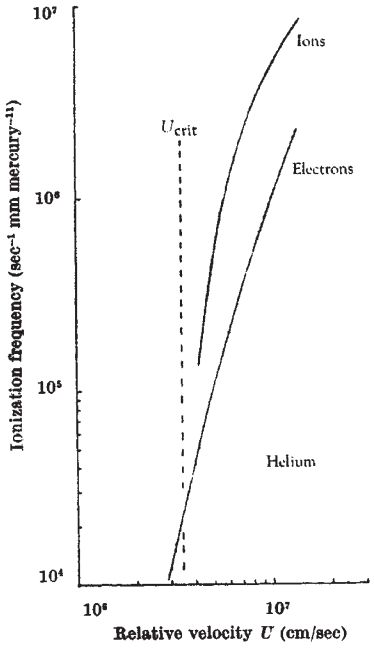


Fig. 1. Plot of ionization frequency against relative velocity U between the plasma and the neutral gas for electrons and ions of helium.

The atoms formed in charge exchange collisions will move in straight lines with almost the same velocity as the ion had before the collision. This means that these atoms will have on average an energy, relative to the gas stream, which is roughly equal to the ion energy. (In the constant mean free time approximation the "new" atom and ion energies are equal[5].) Also the ionization cross-section for atom–atom collisions[10] is similar to that for ion–atom collisions, so that the ionization frequency of the "new" atoms is similar to that of the ions. The "new" atoms may contribute more to the total ionization than the ions, depending on the density of "new" atoms in comparison with that of the ions.

It is suggested that the critical velocity hypothesis is a consequence of the fact that the ionization frequencies of the electrons, ions and atoms formed in charge exchange collisions all become appreciable and increase rapidly with U when U is near U_{crit}. By lending support to the critical velocity hypothesis these results indirectly support Alfvén's theory of the origin of the solar system, but it is pointed out that the details of Alfvén's theory might have to be modified to take account of the possible effects of the ionization caused by the ions and atoms formed by charge exchange.

Results from experiments with partially ionized crossed field homopolar discharges[11,12] can be explained if the relative velocity between the gas and plasma is limited to a value near U_{crit} (ref. 12). If the theory of Drobyshevskii[13] is extended to include the effects of ionization by the ions and atoms formed by charge exchange collisions, fair agreement with experiment is obtained[5]. It turns out that in these experiments the density of "new" atoms is higher than the ion density so that most of the ionization is caused by the atoms formed by charge exchange.

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PLANETARY SCIENCE

Northward Drift of India—Examination of Recent Palaeomagnetic Results

Initial surveys[1–3] of the palaeomagnetism of the Deccan traps showed that there is an apparent decrease of palaeomagnetic inclination with altitude above sea level. This was interpreted[2,4] as indicating a continuous northward movement of India during the time when the traps were being extruded. It was pointed out, however, that the observations could also be explained in terms of partial magnetic instability causing the directions to be pulled towards the present geomagnetic field. Sahasrabudhe[5] has now made a very extensive palaeomagnetic study of the Deccan traps using magnetic cleaning techniques. His aim was to try to use the palaeomagnetic directions for stratigraphic purposes and he did not attempt to

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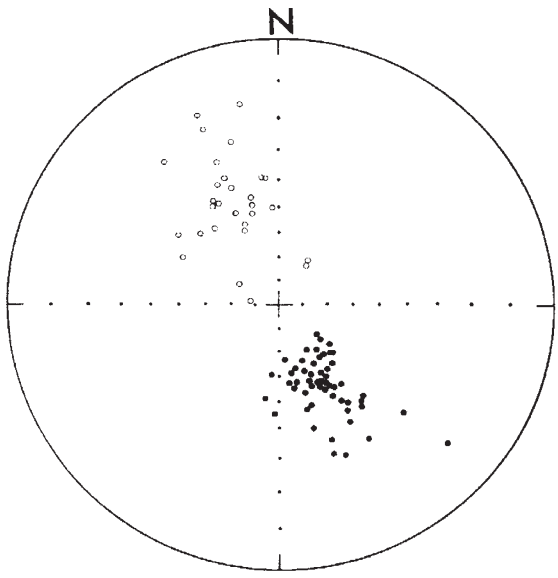


Fig. 1. Stereographic projection of directions of magnetization at ninety-two sites in the Deccan traps. ●, Lower hemisphere; ○, upper hemisphere.

analyse his data for normal palaeomagnetic purposes. I have made a complete analysis of his data and the purpose of this communication is to demonstrate that the earlier results are more likely explained in terms of viscous magnetization. This conclusion leads to some consequences regarding the northward drift of India during the Tertiary.

Approximately a thousand samples were analysed by Sahasrabudhe from seventy-one sites in a minimum of thirty flows and from thirteen dykes intruding them. The sites were spread over the length and breadth of the area at present covered by the traps and over the entire thickness of about 4,000 ft. Flows below an altitude of 2,000 ft. were usually reversely magnetized (positive dips) and those above this level were usually normally magnetized (negative dips). The more detailed collection coupled with magnetic cleaning studies shows that there is no systematic change in inclination through the sequence as was suggested by the earlier results. The lower (reversed) traps range in inclination in no systematic manner from $+70°$ to $+27°$ and the upper (normal) traps likewise range from $-77°$ to $-19°$. Normally magnetized dykes intrude both the lower and upper traps, but reversely magnetized dykes were not found intruding the normal ones. Because the normally magnetized traps always overlie the reversed ones and are therefore stratigraphically younger, a distinction can be made between the lower and upper traps on the basis of polarity and the overall means for the two groups are given in Table 1. This analysis was not carried out by Sahasrabudhe, and for completeness I have also included results from eight sites in four areas not sampled by him[6,7]. In each case, magnetic cleaning techniques were also carried out.

Table 1 shows that ninety-two sites have been sampled in the flows and dykes of the Deccan traps and this is now one of the most thoroughly sampled formations in the palaeomagnetic literature. The means of the two groups differ by $172 \pm 8°$, so that they both represent the same axis of rotation within the experimental errors. The directions of magnetization are illustrated in Fig. 1. The scatter of directions can readily be explained in terms of secular variation[6,9], and in neither case can it be shown[10] that the distribution of directions is non-Fisherian[11]. The data therefore strongly support the contention that little or no polar movement took place during the period of extrusion of the Deccan traps and that the centre of the Indian subcontinent was situated about 30° S. throughout this time.

Recently, some potassium–argon age determinations have been made at the Department of Terrestrial Magnetism, Washington[12]. Although the five determinations range from 42 to 65×10^6 yr, the oldest age is from the overlying Pavagarh basalt, so there is the possibility of argon loss from some of the samples. Assuming the age to be represented by the oldest determination, this places the traps at the Cretaceous–Tertiary boundary[13]. Some clue as to the time covered by the extrusion of the traps is given by the fact that in spite of very detailed sampling only one reversal of the geomagnetic field is recorded in the 4,000 ft. sequence. Recent observations from magnetic anomalies and ocean-floor spreading since the late Cretaceous have led Heirtzler and his colleagues[14] to deduce a possible geomagnetic time scale back to 75×10^6 yr. This analysis suggests that periods of normal or reversed polarity have persisted for no longer than 2×10^6 yr at a time since the late Cretaceous. If this is the case, then it seems unlikely that the Deccan traps cover a period of much longer than 5×10^6 yr.

Polar movement with respect to India since the Upper Permian is illustrated in Fig. 2. It is not clear from the available data whether there is any real difference between the Upper Permian and Jurassic poles, although there is a significant difference between the Jurassic and Cretaceous poles. It is quite clear from Fig. 2, however, that the important polar shift with respect to India occurred between Deccan trap times and the present. During this time the central part of India moved from latitude 30° S. to 20° N. If India had reached its present position by the time of formation of the Himalayas in, say, the Miocene, then this 50° change of latitude took place over about 50×10^6 yr. This corresponds to an

Table 1. PALAEOMAGNETIC RESULTS FROM THE DECCAN TRAPS, INDIA

	No. of site	Directions: Declination	Inclination	Resultant of N unit vector	Estimate of precision (ref. 11)	α_{95}	Pole position: Lat. N.	Long. E.	A_{95}
Upper traps (negative dips)	29	335	−45	27·185	15·4	7	37	102	8
Lower traps (positive dips)	63	152	+53	61·105	32·7	3	30	101	3
Overall mean	92	153	+51	88·114	23·4	3	33	102	3

α_{95} and A_{95} are the radii of 95 per cent circles of confidence.

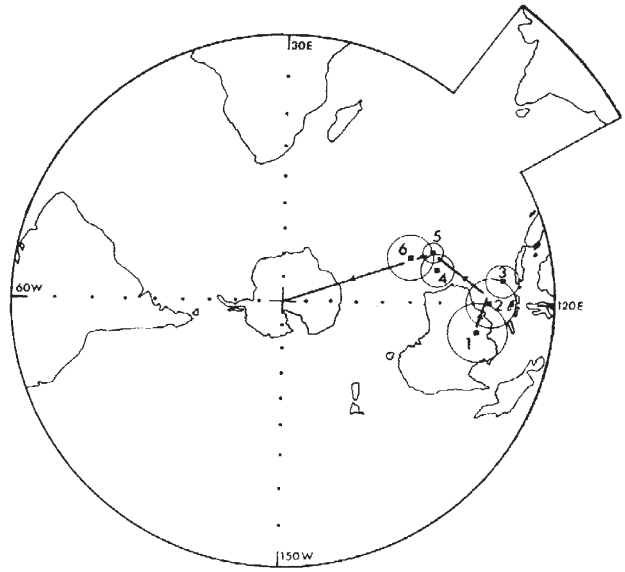


Fig. 2. Phanerozoic polar wander curve for India[15]. Pole numbers refer to Kamthi beds: 1, Upper Permian; 2, Sylhet traps, Jurassic; 3, Rajmahal traps, Jurassic; 4, Tirupati sandstones, Cretaceous; 5, Lower; 6, Upper Deccan traps.

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average rate of drift of India of 10 cm/yr during the Tertiary. This is considerably greater than the average rate of about 5 cm/yr which I suggested[15] occurred between the Gondwanic continents during the break-up of Gondwanaland since the Upper Cretaceous. It will be interesting to see in future whether these palaeomagnetic deductions of drift rates are substantiated by sea floor spreading rates south of India.

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Glasgow University Exploration Society Expedition to Inishtrahull

A Glasgow University Exploration Society expedition to the island of Inishtrahull (lat. 55° 26′ N., long. 7° 14′ W.), which is 5 miles from the northern coast of Donegal, was made during June 23–July 4, 1967, under the leadership of John Smart. The rocks of the island have been both compared and correlated with the Lewisian gneisses of north-western Scotland[1,2], and the geological investigations carried out by the ten members of the expedition, under the direction of Dr D. R. Bowes and Dr A. M. Hopgood, are in continuation of Lewisian investigations being carried out in the Departments of Geology of the University of Glasgow and the University of St. Andrews. The island is approximately 1 mile long, in an E.N.E W.S.W. direction with almost uninterrupted clean coastal exposures on a broad wave-cut platform or on accessible cliffs. Inland elevated parts (up to 141 ft. above sea level at the western end), with outcrops of gneiss and related rocks, occur on either side of a low central area. The island is less than 200 yd. across in part of this central area and the gneiss complex largely covered by post-glacial bedded raised beach deposits (at approximately 40 ft. O.D.) and soil. On the high ground at the western end of the island stands the Inishtrahull Light.

Quartzo-feldspathic and related gneisses make up much of the rock outcrop. Some are strongly foliated and have a well developed mineral lineation; other lithological units show little expression of planar or linear fabric elements. One massive unit of calcium-rich epidote-bearing gneiss has a chemical composition similar to that of a calcareous pelite. Its occurrence among a lithologically layered assemblage of gneisses suggests that the metamorphic complex largely represents a supracrustal series, possibly of sedimentary derivation. There is evidence of metamorphic reconstitution at amphibolite facies, but over large areas the results of retrogressive deformation accompanied by considerable cataclasis constitute the dominant fabric elements of the rocks. Among these crushed units are long narrow zones of cataclasite and small veins of pseudotachylite have been observed.

Amphibolite occurs as small lensoid rock masses and in parts as fragments in agmatite. A narrow, elongate lens of mafic rock with relict pyroxene and a ribbed outcrop, resembling that of layered ultrabasic bodies in the Lewisian of north-western Scotland[3], suggests that there was igneous activity before the development of the dominant foliation. There are minor intrusions, including two sets of pegmatite veins. Their respective times of emplacement in relation to the history of the gneiss complex can be determined from the various effects of metamorphic reconstitution and deformation exhibited. In one case the evidence is based on the presence of an exceptionally well-displayed set of folded joints. A number of unmetamorphosed dolerite dykes with an approximately easterly trend cut the gneiss complex. They are presumed to be of Tertiary age.

The effects of six phases of folding and associated deformation have been observed in the gneiss complex. Of these, the earliest recognized was broadly co-eval with the formation of the gneissose foliation. The structures which exert the dominant control on the distribution of the lithological units, and with which the development of the strong retrogression and cataclasis are related, were formed during the second deformational phase. These structures deform both lithological layering and gneissose foliation. The effects of the third and fourth deformational phases include some new mineral growth as well as the warping of the major structures while the final two phases are only locally expressed as very open folds. In summary the sequence of geological events shown by the gneiss complex is as follows.

(1) The formation of a lithologically layered sequence, possibly as the result of sedimentary processes.

(2) The emplacement of mafic-rich igneous masses.

(3) The development of gneisses associated with tight isoclinal folding of the lithological layers and tectonic disruption, in parts with the formation of agmatite. Metamorphic reconstitution took place and the dominant foliation developed parallel to the axial planes of the isoclinal folds, hence generally parallel to the lithological layering.

(4) The emplacement of basic igneous minor intrusions.

(5) The formation of folds of varying style (isoclinal, asymmetrical, chevron, open) which fold the foliation and exert the most important control on the distribution of the lithological units. Statistical analysis of fabric elements confirms field observations that the folds have sub-vertical axial planes and indicates that they plunge at approximately 15° towards 055°. The style of the structures and the related cataclasis indicate deformation at a relatively shallow level in the crust. Subsequent uprise of quartzo-feldspathic pegmatitic material from deeper levels was controlled by the axial planes of these folds.

(6) Successive emplacement of acid and intermediate minor intrusions at the close of the second deformational phase. Quartzo-feldspathic sheets, much sheared parallel to their margins, are cut by more mafic material which shows some planar alignment of minerals parallel to the intrusion walls.

(7) The development of open folds with sub-vertical axial plane cleavage which trends N.N.E S.S.W. and cuts the minor intrusions of (6). These structures were superimposed on an already deformed foliation, and so the angle of plunge of the folds varies.

(8) The formation of open folds with sub-vertical axial planes which trend N.W.–S.E. These folds are expressed on map scale as warps on the steep limbs of the 55° trending structures and have an associated fine planar mineral growth. In one place a fold of this phase has been observed to deform the axial plane of a fold formed during event (7).

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